

(Applications Development and Emerging Technologies)

TECHNICAL-SUMMATIVE ASSESSMENT

2

PHP ARRAYS AND PRE-DEFINED FUNCTIONS

Student Name / Group Name:	Arianne Julian A. Cruz											
Members (if Group):	<table border="1"><thead><tr><th>Name</th><th>Role</th></tr></thead><tbody><tr><td> </td><td> </td></tr><tr><td> </td><td> </td></tr><tr><td> </td><td> </td></tr><tr><td> </td><td> </td></tr></tbody></table>		Name	Role								
	Name	Role										
Section:	TW25											
Professor:	Prof. Joseph Calleja											

I. PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements.

II. COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Understand and apply best practices and standards in the development of website.

III. INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- To know the different approach of using arrays in PHP.
- To use PHP lazy function foreach to iterate through array elements.
- To implement one dimensional and multi-dimensional in the program.
- To know use variables that is globally declared and locally declared.
- To know how to include separate PHP code in the main page for code enhancement.
- To be familiar with the use of common predefined function such as define, include, and require.
- To use different available mathematical function for manipulating numbers.

IV. BACKGROUND INFORMATION

PHP Arrays and User Defined Functions

Array: one dimensional

Example: Code 1

```
<?php
    $fruit[0] = "orange";
    $fruit[1] = "apple";
    $fruit[2] = "grapes";
    $fruit[3] = "banana";
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Example: Code 2

```
<?php
    $fruit[] = "orange";
    $fruit[] = "apple";
    $fruit[] = "grapes";
    $fruit[] = "banana";
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Example: Code 3

```
<?php
    $fruit = array("orange", "apple", "grapes", "banana");
    echo "<pre>";
    print_r($fruit);
    echo "</pre>";
?>
```

Output:

```
Array
(
    [0] => orange
    [1] => apple
    [2] => grapes
    [3] => banana
)
```

PHP Predefined Functions

Using Constant

define() functions

used to declare constants. A constant can only be assigned a scalar value, like a string or a number. A constant's value cannot be changed.

Syntax:

```
define('NAME', 'value');
```

Example:

```
<?php
    define('MAX_VALUE', 10);
    for($i=5; $i<MAX_VALUE; $i++){
        echo $i." ";
    }
    //MAX_VALUE = 20; is invalids
?>
```

Output:

```
5 6 7 8 9
```

V. GRADING SYSTEM / RUBRIC (please see separate sheet)

VI. LABORATORY ACTIVITY

1. **Using ARRAYS**, create a fruit directory that will show the image, name, description and facts about the fruit (create at least 10 fruits sort in alphabetical order). Implement a HTML and CSS code for the structure and design of the webpage. Please see the sample output below.

My Fruits			
Image	Name	Description	Facts
	Banana	Color Yellow	Bananas are a healthful addition to a balanced diet, as they provide a range of vital nutrients and are a good source of fiber.

2. **USING USER DEFINED FUNCTION.** Create a formula for each volume of shapes (Cube, Right Rectangular Prism, Prism or Cylinder, Pyramid or Cone and Sphere) then display the result. Please see the sample below:

Volume of Shapes		
Values	Formula	Answer
$s = 5$	$V=s^3$	125

3. Convert your Student Resume into a web based form, used include() and require() functions to connect all pages

Include and create the following pages:

- Career Objective page
- Personal information page
- Educational Attainment page
- Skills page
- Affiliation page
- Work Experience Page

Please see the sample below:

	Personal information
• Career Objective	
• Educational Attainment page	
• Skills page	
• Affiliation page	
• Work Experience Page	

VII. QUESTION AND ANSWER

1. What is Array?

An array is a data structure which uses a single variable to store multiple values. It enables efficient indexing and accessing of its elements or keys. Arrays are the basic data structure used to store data in collections in programming.

2. What is function in PHP?

Functions are code segments that can be reused in PHP to carry out tasks in programming. The code segment receives inputs, manipulates the information, and generates output. Functions make code manageable and reduce code repetition.

3. Differentiate the one-dimensional array to multi-dimensional array

In a one-dimensional array, you have only one list of items that can be accessed using a single index value. In a multidimensional array, arrays are nested in each other, hence you will need more than one index to access them.

4. Describe the difference between Global Variable to Local Variable

Global variables can be accessed from anywhere in the program. Local variables can only be used within the particular function or block in which they have been defined. By using local variables, we avoid any accidental side effects and make our code safer.

5. What is the difference between function without parameter and a function with parameter?

A function with no parameters works with preset values or global variables, without any outside input. On the other hand, a function with parameters works with specified values that define its behavior.

6. What are the different common predefined functions? Describe each and give example

Some of the commonly used pre-defined functions are `strlen()`, which is used to find the length of strings; `count()`, used to count elements in an array; and `date()`, used to format time. Using these functions does not require any user coding as the solution is already available. For instance, `strlen('Hello')` will return the integer value of 5.

7. Is it useful to use an include or require function on a webpage? Why?

The use of `include` or `require` would help in the modularization of code and in ensuring consistency throughout. This makes it easier to reuse any header or footer on different pages. Changes in one file will automatically update all other included files.

8. What is the importance of a function?

Why are functions essential? Functions encourage code reuse and code modularity. They decompose big tasks into small chunks that are easier to manage, making the code easy to read and debug.

9. What is the importance of a string function?

These functions are highly important for any manipulation and analysis of textual data. With their help, one can perform such actions as searching, replacement, and formatting of text without putting much effort. Their correct usage will also ensure data consistency.

VIII. REFERENCES

1. <https://www.w3schools.com/css/>
2. <https://www.w3schools.com/html/>
3. https://www.w3schools.com/php/php_variables.asp
4. https://www.w3schools.com/php/php_arrays.asp
5. https://www.w3schools.com/php/php_arrays_indexed.asp
6. https://www.w3schools.com/php/php_arrays_associative.asp
7. https://www.w3schools.com/php/php_arrays_multidimensional.asp
8. https://www.w3schools.com/php/php_arrays_sort.asp
9. https://www.w3schools.com/php/php_functions.asp
10. https://www.w3schools.com/php/php_constants.asp
11. https://www.w3schools.com/php/php_includes.asp
12. https://www.w3schools.com/php/php_string.asp

Note: The following rubrics/metrics will be used to grade students' output.

Program (100 pts.)	(Excellent)	(Good)	(Fair)	(Poor)
Program execution (20pts)	Program executes correctly with no syntax or runtime errors (18-20pts)	Program executes with less than 3 errors (15-17pts)	Program executes with more than 3 errors (12-14pts)	Program does not execute (10-11pts)
Correct output (20pts)	Program displays correct output with no errors (18-20pts)	Output has minor errors (15-17pts)	Output has multiple errors (12-14pts)	Output is incorrect (10-11pts)
Design of output (10pts)	Program displays more than expected (10pts)	Program displays minimally expected output (8-9pts)	Program does not display the required output (6-7pts)	Output is poorly designed (5pts)
Design of logic (20pts)	Program is logically well designed (18-20pts)	Program has slight logic errors that do no significantly affect the results (15-17pts)	Program has significant logic errors (3-5pts)	Program is incorrect (10-11pts)
Standards (20pts)	Program code is stylistically well designed (18-20pts)	Few inappropriate design choices (i.e. poor variable names, improper indentation) (15-17pts)	Several inappropriate design choices (i.e. poor variable names, improper indentation) (12-14pts)	Program is poorly written (10-11pts)
Delivery (10pts)	The program was delivered on time. (10pts)	The program was delivered a day after the deadline. (8-9pts)	The program was delivered two days after the deadline. (6-7pts)	The program was delivered more than two days after the deadline. (5pts)